



CHAPTER

38 Judgment Under Uncertainty

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Abstract

This chapter traces a path through over 50 years worth of research on judgment under uncertainty. The literature is viewed through the lens of two criteria for appraising judgment. The first, correspondence, emphasizes the empirical accuracy of judgment; the second, coherence, emphasizes internal consistency and rationality. Major findings from research traditions aligned with these criteria are reviewed (e.g., correspondence: fast and frugal heuristics; coherence: heuristics and biases), and a case is made for viewing the criteria as complementary rather than contrasting. The importance of considering the role of the environment and the mind in understanding judgment is emphasized, and the potential for integrating correspondence and coherence in a causal Bayesian framework is outlined briefly.

Keywords: judgment, uncertainty, coherence, correspondence, rationality, heuristics, biases, fast and frugal heuristics, lens model, dual systems, Bayes, causal models

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As we know, there are known knowns. There are things we know we know. We also know there are known unknowns. That is to say we know there are some things we do not know. But there are also unknown unknowns, the ones we don't know we don't know.

—Donald Rumsfeld

Donald Rumsfeld's quote neatly illustrates the fact that uncertainty is a pervasive feature of our lives. Many decisions from the momentous to the mundane are based on our beliefs about the chances of future events. Should we go to war against a country? It depends on the chance that the war will be a "success." Should we take an umbrella when we leave for work in the morning? It depends on the chance of rain during the day. This chapter focuses on how people assess the probability of events, make judgments and predictions, and cope with living in an inherently uncertain world. The chapter proceeds by first considering how we can evaluate the quality of judgment made under uncertainty and then traces a path through research programs that have aligned themselves with two different ways of appraising judgments: via coherence and correspondence.

How do we know if a judgment is good or bad? If we take the umbrella to work and it rains, then our judgment about the weather would be considered good. If it does not rain, then our judgment would be considered bad. Thus, in this situation the quality of our judgment is determined by whether what we predicted to happen did, in fact, occur. That is, whether our judgment *corresponded* with the eventual state of the world. Hammond (1996) contrasted *correspondence* with *coherence*—criteria that are concerned not with the accuracy of predicting events, but with the *rationality* of a judgment. Hammond describes the two approaches as follows:

Correspondence theory focuses on the empirical accuracy of judgments, irrespective of whether the cognitive activity of the judge can be justified or even described.

Coherence theorists...examine the question of whether an individual's judgment processes meet the test of rationality—internal consistency—irrespective of whether the judgment is empirically accurate. (Hammond, 1996, p. 106)

The emphasis on the empirical accuracy of judgments in correspondence theory has led researchers to focus on the ability of judges in many real-world environments to predict the outcome of uncertain events (e.g., the chance of a patient dying given a particular diagnosis). In contrast, coherence theorists have tended to focus on decontextualized problems, which lend themselves to being solved by normative rules, such as the laws of probability (e.g., predicting the outcomes of successive coin flips). Although these criteria are often presented as conflicting (Dunwoody, 2009; see also Gigerenzer, 1996; Kahneman & Tversky, 1996), it is perhaps more useful to see them as complementary. The different foci of the two approaches elucidate different aspects of judgment behavior that together can help place our understanding of human judgment into a general cognitive framework (see Newell, Lagnado, & Shanks, 2007). This chapter develops this view by exploring the synergies between the approaches and concludes by advocating a potential new path in which the criteria can be combined. The next section introduces the correspondence-based approach, which is primarily concerned with quantitative judgment (or prediction) under uncertainty.

Correspondence-Based Approaches

Egon Brunswik's research on visual perception (e.g., Brunswik, 1956) is often credited as inspiring the correspondence approach to human judgment (Dunwoody, 2009). The main ideas in the Brunswikian approach are that an object in the environment (a "distal" stimulus) produces multiple cues through the stimulation of the perceiver's sense organs. These "proximal" cues are necessarily fallible (due to the probabilistic nature of the relation between the cues and the environment) and therefore only imperfectly indicate the true state of the external environment. Thus, perception is a constructive process, involving inferences drawn on the basis of incomplete and ambiguous sensory information. Hammond's (1955) important contribution was to demonstrate that the principles of perception proposed by Brunswik could be applied to the study of judgment.

Beginning with clinical judgment, Hammond and his colleagues demonstrated that Social Judgment Theory, as it became known, could be applied to a wide range of situations involving judgment. The main "tool" of the social judgment theorist is the Lens Model: a metaphor for thinking about how a "to-be-judged" criterion in the world (e.g., whether a patient is psychotic or neurotic) relates to the judgment made in the "mind" of the judge, that is, how well the judgment *corresponds* with the event in the world. The "lens" captures the notion that the judge views the world through an imperfect set (or lens) of cues.

Karelaia and Hogarth (2008) conducted a meta-analysis on over 240 different lens-model studies comprising both laboratory and field studies of judgment and drew three major conclusions: (1) that people are capable of achieving high levels of judgmental performance—as indicated by an average "achievement" correlation between the environmental criterion and the judgment of .56; (2) in studies where feedback was provided people learned best when the feedback instructed them about the characteristics of the task (e.g., how much "weight" a particular cue should be given in forming a judgment); and (3) that *models* of people's judgments tend to be more accurate than people's own judgments. The first conclusion is heartening, the

second perhaps not overly surprising, but the third seems counterintuitive and a little worrying. Why should a model of people's judgments outperform their own judgments?

Judgments Made by Humans and Models

Meehl (1954) demonstrated that judgments made by experts—usually clinicians—were often inferior in terms of accuracy to simple statistical models provided with the same information. Two classes of model can be identified—one in which information from the environment is fed into a statistical model, and one in which information derived from the judgments made by a person (typically an expert) is used.

p. 605 To illustrate these techniques, consider the following example, adapted from Baron (2008). Imagine that you wanted to predict the performance in a final exam from students' performance on three pieces of assessment (A, B, and C) completed during the term. A statistical technique for prediction is to conduct a multiple linear regression—this simply finds a linear sum of the predictors (the assessment scores) in which the weight assigned to each assessment minimizes the error between the predicted and actual final exam performance. For example, the model might settle on ideal weights of .3 for assessment A and B, and .4 for assessment C. Provided the predictors have a linear relationship with the criterion, as we would expect to occur here, such a technique can work well.

The second technique is to give the same information to a judge (e.g., the professor teaching the course) and ask him or her to make a prediction about the final exam scores. The inputs would be the same—the assessment scores—but the weights and the rule for combining the inputs might be different. For example, the professor might believe that assessment B is a better indicator of final performance than assessment A. By analyzing the judgments the professor makes about several students, it is possible to develop a “model” of the judge that captures the way in which he or she weights the cues. Once we have this model we do not need the judge anymore but can simply “plug in” the assessment scores for other students and make predictions.

The superiority of these statistical techniques has been corroborated by hundreds of studies in diverse contexts (e.g., Dawes, Faust, & Meehl, 1989; Grove, Zald, Lebow, Snitz, & Nelson, 2000), leading some researchers to conclude, “Whenever possible, human judges should be replaced by simple linear models” (Hastie & Dawes, 2001, p. 63).

The Advantages of Statistical Models

Dawes et al. (1989) list several factors that can contribute to the superior performance of statistical methods. Two key factors are, first, a statistical method will always arrive at the same judgment for a given set of data. Experts, on the other hand, are susceptible to the effects of fatigue, information presentation (framing), and recent experience. Second, experts are often exposed to a skewed sample of evidence, making it difficult to assess the actual relation between variables and a criterion of interest. This means that there is little opportunity to obtain the “task information” feedback that Karelaia and Hogarth (2008) identified as being so important for improving judgment.

There are, however, times when a model will not prove superior; in particular, when the idiosyncratic knowledge of a judge is not captured by a linear model. Imagine, for example that the professor knew that a particular student had been sick during the term and had underperformed on her assessments. This information could be included by the professor to modify his prediction, but the model, with only the scores as input, would miss this subtlety. However, Karelaia and Hogarth (2008) conclude that, these subtleties notwithstanding, “the inconsistency that people exhibit in making judgments is sufficient for models of their judgments to be more accurate than they are themselves” (p. 420).

In summary, studies in the lens model tradition have emphasized that human quantitative judgment can show relatively good correspondence with the environment (e.g., the overall achievement correlation of .56; Karelaia & Hogarth, 2008) but that models, which are unaffected by the vagaries of human cognition, show better correspondence. Thus, human judges are not superfluous because they still serve an important role as the provider of information for the models of their judgment, but models should be used to weight and integrate this information (Baron, 2008). The next section considers how the vagaries of cognition impact on a different type of judgment under uncertainty: probability judgment.

The coherence tradition investigates whether our judgments follow or *cohere* with normative theory. For example, when we check the weather in the morning, our judgment about the chance that it will rain and the chance that it will not must be complements of one another. If we believe that there is a 30% chance of rain, then to maintain a coherent set of beliefs, we must also believe that there is a 70% chance of no rain; believing otherwise would violate the laws of probability.

p. 606

These laws were developed in the 17th century (Hacking, 1975) and can be thought of as a set of rules or axioms that should be followed to ensure sound judgment and reasoning (e.g., Ramsey, 1931). Failure to cohere to these laws leads to poor or defective judgments because, among other things, it allows unscrupulous individuals to take advantage of your incoherency. The best example of this vulnerability is the so-called Dutch book. This is a situation in which your probability judgments lead you to accept bets in which you are bound to lose no matter how the events you bet on turn out (see Newell et al., 2007, Chapter 5 for a worked example).

Coherent beliefs by themselves, however, are not sufficient to ensure the truth of a judgment. We could be entirely coherent in judging the probability of the moon being made out of cheese as .8, provided we also judged the probability that it was not as .2. Thus, although coherence and correspondence are often cast as contrasting, they are, in fact, inescapably entwined. Dunwoody (2009) makes this point cogently: “If nature is necessarily lacking in self-contradictions, as the coherence theorists argue, then beliefs that correspond with nature should ultimately be coherent as well” (Dunwoody, 2009, p. 117). This idea accords with the thesis of this chapter that both criteria need to be considered in appraising judgment. Nonetheless, many studies have examined coherence in “isolation,” as the next section illustrates.

Bayes's Theorem and Conservatism

Put simply, Bayes's theorem is a method for working out what we do not know from what we do know. The theorem tells us how we should update our belief about an event or hypothesis in light of new evidence relevant to the hypothesis. Consider the following example taken from Edwards (1968):

In front of you are two bags each containing 100 poker chips. You know that one bag has 70% red and 30% blue chips, and the other has 30% red and 70% blue. One bag is chosen at random (coin flip). Eight red and four blue chips are then drawn with replacement, in 12 consecutive drawings. What is the probability that the bag the chips are drawn from is the predominantly red bag?

To answer the question, you need to consider the prior odds of the bag being the predominantly red one (i.e., the strength of your belief before any chips had been drawn), and multiply this by the likelihood ratio of the data (the information provided by the sampled chips) to get the posterior odds (your revised belief). The prior odds are clearly .5 because the bag was chosen via a coin flip, but what are the correct posterior odds? Ward Edwards and colleagues (e.g., Edwards, 1968; Phillips & Edwards, 1966) gave problems like this to participants to see whether the difference between their prior and posterior odds was in accord with that prescribed by Bayes's theorem. In the example, appropriate belief revision leads to posterior odds of .97 for the selected bag being the predominantly red one (see Sedlmeier, 1999 for a worked example if you find this answer difficult to fathom!). However, Edwards and colleagues found that the difference in probability estimates provided by participants was typically too small. People tended not to “extract from the data as much certainty as is theoretically implied by Bayes's Theorem” (Phillips & Edwards, 1966, p. 346)—an effect labeled *conservatism*.

Faced with this departure from the predictions of the theorem, Edwards's solution was to modify the theorem to account for the data rather than abandoning the model as an inappropriate benchmark. This approach was characteristic of much of the literature on human judgment at that time (e.g., Peterson & Beach, 1967; see also McKenzie, 2005). However, the view that normative models provided good approximations for psychological theories of judgment under uncertainty was about to be challenged.

Writing in 1983, Ward Edwards lamented that his work on conservatism in probabilistic judgment had led to a “spate of studies that purport to show that men are incompetent intellectually” (p. 216). The comment was a reaction to the now seminal work of Tversky and Kahneman (1974), which focussed on the *heuristics* that people adopted in making judgments under uncertainty and the predictable *biases* that resulted from their use. This work led to a reversal of the optimistic way in which departures from the predictions of normative models had been interpreted. Now such failures of coherence were taken, by some at least, to paint the picture of man as a “cognitive cripple.” Although this was not Tversky and Kahneman’s intended message, as the following section illustrates, it is easy to see how one might be left with this pessimistic impression.

The heuristics and biases approach can best be explained via appeal to two related concepts: attribute substitution and natural assessment (Kahneman & Frederick, 2002). Attribute substitution refers to the idea that when people are asked to make a judgment about a specific *target attribute* (e.g., How probable is X?), they instead make a judgment about a *heuristic attribute* (e.g., How representative is X?), which they find easier to answer. This ease of answering often arises because the heuristic attributes relied upon are readily accessible via the “natural assessment” of properties such as size, distance, similarity, cognitive fluency, causal propensity, and affective valence. Kahneman and Frederick (2002; ↪ see also Tversky & Kahneman, 1983) argue that because such properties are “routinely evaluated as part of perception and comprehension” (p. 55) they come more easily to mind than the often *inaccessible* target attributes. Hence, target attributes are *substituted* for heuristic attributes.

The evidence for attribute substitution and natural assessment comes in two main guises. The first is the simple demonstration that judgments of target attributes correlate very highly with judgments of the heuristic attributes with which they have been substituted. The second is that the failure to cohere with normative laws lead, in certain situations, to striking biases. There is an abundance of evidence documenting various heuristics and biases (see the volumes by Kahneman, Slovic, & Tversky, 1982 and Gilovich, Griffin, & Kahneman, 2002 for examples); one of the most famous demonstrations is the so-called Linda problem.

The Linda Problem: Representativeness as a Heuristic Attribute

Tversky and Kahneman (1982) gave participants the following simple personality sketch:

Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.

They then asked separate groups of participants to rank a set of eight statements about Linda either by how *representative* they appeared to be or how *probable* they were. The correlation between the rankings was .99. Tversky and Kahneman took this as very powerful evidence that when people were asked a question about probability, they substituted this target attribute with a heuristic one about representativeness; that is, the degree to which one “object” (Linda in the personality sketch) resembles the “object” of the statement (e.g., Linda is a psychiatric social worker).

However, there was an additional “experimental flourish” (Kahneman & Frederick, 2002) that provided the dramatic demonstration of judgment via representative thinking. In another version of the problem (Tversky & Kahneman, 1983), participants were asked: Which of these two statements is more probable? (1) Linda is a bank teller; (2) Linda is a bank teller and active in the feminist movement. The overwhelming majority response was to rank (2) as more probable than (1). This is a clear violation of the conjunction rule of probability: A conjunction cannot be more probable than either of its conjuncts. All feminist bank tellers are, by definition, bank tellers so a person cannot be *more likely* to be a *feminist* bank teller than just a bank teller. Nonetheless, the description of Linda is highly representative of active feminists but not of bank tellers; thus, a judgment by representativeness leads to statement (2) receiving the higher ranking.

Representativeness-based explanations have also been applied to a related class of problems in which participants appear to neglect base-rate frequency information and rely on specific details of scenarios. In the well-known “lawyers and engineers” problem, participants were shown brief personality sketches of

several individuals allegedly sampled at random from a group of 100 lawyers and engineers. They were then asked to assess for each one the likelihood that the person was an engineer rather than a lawyer. In one condition participants were told that the group comprised 70 lawyers and 30 engineers; in the other condition the composition was reversed. Bayes's rule dictates that the odds of any particular individual being an engineer are 5.44 times higher in the second condition than the first. In striking contradiction of the rule, participants' judgments in the two conditions were remarkably similar. Tversky and Kahneman (1974) argued that the judgments were made "by the degree to which this description was representative of the two stereotypes with little or no regard to the prior probabilities of the two categories" (p. 1125).

Availability as a Heuristic Attribute

p. 608

Although the heuristics and biases approach is typically associated with departures from coherence criteria, this focus is not exclusive (cf. Dunwoody, 2009). Availability, a pervasive heuristic attribute, has been invoked to explain errors of correspondence. Availability has been described both as a measure of the ease with which instances come to mind (Kahneman & Tversky, 1996) and the number of instances that come to mind (Tversky & Kahneman, 1973). Most of the time this heuristic will serve us well as a substitute for estimating frequencies or probabilities—typically more likely events come to mind more readily. However, at times the heuristic will produce errors, and it is these errors that have been the focus of research. Take, for example, a judgment of whether homicide or suicide is more likely in the US adult male population. The majority response is homicide, although in fact suicides are far more common (Lichtenstein, Slovic, Fischhoff, Layman, & Coombs, 1978). The explanation for this error is simply that instances of homicide come more readily to mind (are more available) than those of suicide (perhaps due to a bias in media coverage).

Note that the biased judgment of homicide arises because of the way in which information is presented to us from external sources (e.g., the media), but availability can also sometimes be a poor guide when we search through information in our memories. Consider this example: What is more probable: that a word starts with the letter *k* or has *k* as its third letter? Words with *k* as the third letter are in fact far more common, but people tend not to give this answer. Tversky and Kahneman (1973) argued that because we tend to organize words by their initial letters—and thus search our memories for them in this way—we tend to think that words starting with *k* occur more frequently.

Affect as a Heuristic Attribute

The notion of an "affect heuristic" is strongly tied to availability. Indeed, the only difference between the two seems to be that the former is invoked in situations involving affectively loaded responses. Kahneman and Frederick (2002) point out that the affective valence of a stimulus is something that is often "naturally assessed" (we cannot avoid our affective reaction to many stimuli—disgust, desire, etc). Thus, affect is an obvious candidate for a heuristic attribute. According to Finucane, Peters, and Slovic (2003), the basic tenet of the affect heuristic is that "positive and negative affective feelings, attached to images, guide and direct judgments and decisions" (p. 341). In essence, the affect heuristic works via the generation of instances or images, which are "tagged" with particular feelings. These feelings or "affects" are then used as substitutes for whatever target attribute is being interrogated.

To test this idea, Finucane, Alhakami, Slovic, and Johnson (2000) presented participants with vignettes designed to manipulate affect by describing either the benefits or risks of nuclear power. They then collected perceived risk and benefit ratings. The general pattern in the results was that providing information about one attribute (e.g., risk) had a carryover effect on the attribute about which nothing had been learned directly (e.g., benefit). Finucane et al. interpreted this pattern in terms of people "consulting their overall affective evaluation of the item when judging its risk and benefit" (p. 13)—in other words, people relied on an affect heuristic to make risk/benefit judgments.

The notion that affect influences judgment is also supported by work examining the effect of different ways of presenting statistical information. Slovic, Monahan, and MacGregor (2000) demonstrated that clinicians provided with recidivism risks presented as frequencies (e.g., 20 out of 100) judged mental patients as posing higher risks than when the same information was presented as probabilities (e.g., 20%). The explanation was that only the frequency presentation generated a "terrifying image" of the recidivist in the mind of the clinician and that the affect associated with this imagery led to the more extreme judgments. In a similar vein, Newell, Mitchell, and Hayes (2008) demonstrated that participants were more willing to take

part in positive activities (e.g., playing a lottery) but less willing to take part in negative activities (e.g., complex laser surgery) when frequencies were used to communicate the chances/risks involved than when probabilities were used.

Anchoring and Adjustment

Anchoring—the tendency to insufficiently adjust from an initial value when estimating continuous variables (e.g., age, height, weight, price, etc.)—was introduced as one of the three major heuristics (alongside availability and representativeness) by Tversky and Kahneman (1974). However, Kahneman and Frederick (2002) note that anchoring does not fit within the attribute substitution framework. Rather, the heuristic works by increasing the plausibility of a particular value of the target attribute. The typical task involves providing a high (or low) anchor to separate groups of participants (e.g., Do you think Gandhi died before or after the age of 140 (9) years old?) and then asking for an estimate (e.g., How old was Gandhi when he died?). The highly robust finding is that those given the high anchor will estimate a higher value (67 years old, in this case) than those given a low anchor (50 years old) (Strack & Mussweiler 1997). Thus, although the initial value is irrelevant, it tends to be assimilated into the provided estimate.

Critiques of the Heuristics and Biases Approach

p. 609 The heuristics and biases program has been highly influential; however, some have argued that in recent years the influence, at least in psychology, has waned (McKenzie, 2005). This waning has been due in part to pointed critiques of the approach (e.g., Gigerenzer, 1996). The critique comprises two main arguments: (1) that by focussing mainly on coherence standards the approach ignores the role played by the environment or the context in which a judgment is made; and (2) that the explanations of phenomena via one-word labels such as availability, anchoring, and representativeness are vague, insufficient, and say nothing about the processes underlying judgment (see Kahneman, 2003; Kahneman & Tversky, 1996 for responses to this critique).

The accuracy of some of the heuristics proposed by Tversky and Kahneman can be compared to correspondence criteria (availability and anchoring). Thus, arguing that the tradition only uses the “narrow norms” (Gigerenzer, 1996) of coherence criteria is not strictly accurate (cf. Dunwoody, 2009). Nonetheless, responses in famous examples like the Linda problem can be reinterpreted as sensible rather than erroneous if one uses conversational or pragmatic norms rather than those derived from probability theory (Hilton, 1995). For example, Hertwig, Benz, and Krauss (2008) asked participants which of the following two statements is more probable:

[X] The percentage of adolescent smokers in Germany decreases at least 15% from current levels by September 1, 2003.

[X&Y] The tobacco tax in Germany is increased by 5 cents per cigarette and the percentage of adolescent smokers in Germany decreases at least 15% from current levels by September 1, 2003.

According to the conjunction rule, X cannot be more probable than X&Y and yet the majority of participants ranked the statements in that order. However, when subsequently asked to rank order four statements in order of how well each one described their understanding of X&Y, there was an overwhelming tendency to rank statements like “X and therefore Y” or “X and X is the cause for Y” higher than the simple conjunction “X and Y.” Moreover, the minority of participants who did not commit the conjunction fallacy in the first judgment showed internal coherence by ranking “X and Y” as best describing their understanding in the second judgment. These results suggest that people adopt a causal understanding of the statements, in essence ranking the probability of X, given Y as more probable than X occurring alone. If so, then arguably their conjunction “error” is no longer incorrect. (See Moro, 2009 for extensive discussion of the reasons underlying the conjunction fallacy, including why “misunderstanding” cannot explain all instances of the fallacy.)

The “vagueness” argument can be illustrated by considering two related phenomena: the gambler’s fallacy and the hot-hand (Gigerenzer & Brighton, 2009). The gambler’s fallacy is the tendency for people to predict the opposite outcome after a run of the same outcome (e.g., predicting heads after a run of tails when flipping a fair coin); the hot-hand, in contrast, is the tendency to predict a run will continue (e.g., a player making a shot in basketball after a succession of baskets; Gilovich, Vallone, & Tversky, 1985). Ayton and

Fischer (2004) pointed out that although these two behaviors are opposite—ending or continuing runs—they have both been explained via the label “representativeness.” In both cases a faulty concept of randomness leads people to expect short sections of a sequence to be “representative” of their generating process. In the case of the coin, people believe (erroneously) that long runs should not occur, so the opposite outcome is predicted; for the player, the presence of long runs rules out a random process so a continuation is predicted (Gilovich et al., 1985). The “representativeness” explanation is therefore incomplete without specifying a priori which of the opposing prior expectations will result. More important, representativeness alone does not explain *why* people have the misconception that random sequences should exhibit local representativeness when in reality they do not (Ayton & Fischer, 2004).

Gigerenzer (1996) argued that for progress to be made in understanding judgment, the field needed to abandon coherence criteria, return to a focus on correspondence, and move away from vague labels of heuristics. The adaptive toolbox of fast and frugal heuristics borne from this approach is reviewed in the next section.

Fast and Frugal Heuristics

At the heart of the fast and frugal approach lies the notion of *ecological rationality*. Ecological rationality is a type of correspondence criterion that emphasizes “the structure of environments, the structure of heuristics, and the match between them” (Gigerenzer, Todd, & ABC Research Group, 1999, p. 18). Thus, the concern is not so much with whether a heuristic is inherently accurate or biased, but rather why and when a heuristic will fail or succeed (Gigerenzer & Brighton, 2009). This focus on the “ecology,” echoing the traditions of Brunswik’s functionalism (1956) and Simon’s bounded rationality (1955), brings context to the fore and overcomes the perceived “cognition in a vacuum” criticism of the heuristics and biases approach (McKenzie, 2005). The other central pillar of the approach is an emphasis on detailed models of the processes underlying each heuristic.

p. 610

Building Blocks and the Adaptive Toolbox

The fundamental building blocks of simple heuristics are rules for *searching* for information, *stopping* search, and *deciding*. Different types of search, stopping, and decision rules are combined into judgment heuristics for particular environments. The metaphor of the mind as an *adaptive toolbox* captures the notion that these heuristics are contained within a central resource from which tools are selected depending on the constraints of the environment and the goals of the decision maker. The emphasis is on using heuristics that do well, rapidly, and on the basis of a small amount of information. The following example serves to illustrate the approach.

Which US city has more inhabitants: San Diego or San Antonio? Goldstein and Gigerenzer (2002) posed this question to groups of students from the University of Chicago and the University of Munich. Sixty-two percent of University of Chicago students inferred correctly that San Diego was larger, but surprisingly, every single Munich university student answered correctly. Goldstein and Gigerenzer explained the result through the operation of the *recognition heuristic*, which states that when you are faced with two objects (e.g., cities) and you have heard of one but not the other, you should choose the former. Most of the Chicago students had heard of both cities so could not rely on this heuristic; in contrast, the ignorance of the Munich students (very few had heard of San Antonio) facilitated their judgment (a “less is more” effect; cf. Gigerenzer & Brighton, 2009). Thus, the recognition heuristic can be described in simple process rules: *Search* your memory for both objects; *stop search* if only one object is recognized; *choose* the recognized object.

If both objects are recognized, then a second heuristic from the toolbox is invoked: *take the best* (TTB). Here, search in memory (or the experimental environment) continues until a cue that discriminates between the two options is found. For example, you might recognize both San Diego and San Antonio but then recall that of the pair only San Diego has a world-famous zoo and infer that San Diego is larger. More formally, TTB states that cues are *searched* in descending order of their predictive validity of the criterion (in this case, city size), *stops* when a single discriminating cue is found, and *chooses* the object to which that cue “points.”

The principle of ecological rationality emphasizes that these heuristics will only be successful in certain types of environments. The recognition heuristic will only work when the criterion of interest is correlated

with recognition. Typically cities that are larger are more likely to be well known, but this is not always the case (cf. Oppenheimer, 2003). Likewise, TTB will be most successful in environments with sharply decreasing cue validities or weights—a so-called noncompensatory environment. In these environments, cues lower in validity, which are “ignored” by TTB, cannot outweigh the evidence provided by the “best” discriminating cue. In contrast, when cue weights are compensatory—all the pieces of information available have approximately equal validity—then TTB will suffer because it will ignore important information (see Hogarth & Karelaia, 2005).

Empirical Evidence for the Use of Fast and Frugal Heuristics

Like the heuristics and biases tradition that preceded it, the fast and frugal approach has inspired a wealth of studies. This research provides clear evidence that in certain environments simple strategies like TTB can be as effective and less resource intensive than more complex ones, such as multiple regression (e.g., Gigerenzer & Goldstein, 1996); that relying on simple name recognition can be very useful, for example, it performs just as well as the ATP rankings in predicting winners of Wimbledon tennis matches (e.g., Serwe & Frings, 2006); and that participants’ behavior in experimental judgment tasks sometimes accords with the predictions of simple strategies (e.g., Bergert & Nosofsky, 2007; Bröder & Schiffer, 2003; Rieskamp & Otto, 2006) and at other times does not (Hilbig & Pohl, 2009; Newell, 2005; Newell & Fernandez, 2006; Newell & Shanks, 2003, 2004; Newell, Weston, & Shanks, 2003; Richter & Spath, 2006).

Perhaps most contentious is whether the fast and frugal approach provides a different class of explanation to that of the heuristics and biases tradition (e.g., Chater, Oaksford, Nakisa, & Redington, 2003; Dougherty, Thomas, & Franco-Watkins, 2008; Newell, 2005). There is no doubt that the renewed emphasis on the role of the environment, the attempt to understand the interface between the mind and the environment (cf. Fiedler & Juslin, 2006), and the exploration of how behavior is adaptive and adaptable represents important progress (e.g., McKenzie, 2005; see also Anderson, 1990; Oaksford & Chater, 1998). However, the extent to which behavior described in terms of the selection and implementation of heuristics from a toolbox is any more or less vague than behavior described in terms of natural assessment and attribute substitution remains debatable, as outlined in the next section.

Critiques of Fast and Frugal Heuristics

The appeal of fast and frugal heuristics lies in their simplicity, precision, and testability. However, once we scratch beneath the surface of this appealing veneer, each of these qualities becomes somewhat tarnished. The simplicity of the heuristics belies a large amount of precomputation or prior learning/knowledge of the validities of particular cues in particular environments (Dougherty, Franco-Watkins, & Thomas, 2008; Juslin & Persson, 2002; Newell, 2005). Although a model like TTB can predict judgments as well as multiple regression (e.g., Gigerenzer & Goldstein, 1996), one of the principal reasons it does so is that TTB is provided with a validity-ordered hierarchy of all of the important cues for a given environment. Thus, the “simple” rules for search, stopping, and deciding are dependent on much more complex (and ill-specified or understood) mechanisms for determining cue hierarchies. Indeed, many researchers have acknowledged and demonstrated the difficulty that participants exhibit in learning the kind of cue-validity hierarchies necessary for implementing TTB (e.g., Bergert & Nosofsky, 2007; Rakow, Newell, Fayers, & Hersby, 2005).

One way of responding to this issue is to suggest that cue hierarchies might be constructed in ways other than individual learning. For example, Gigerenzer, Hoffrage, and Goldstein (2008) suggest that cues might be learned via evolution, or by social learning; the latter simply amounting to being told what cues are important for a given judgment (e.g., diagnosing acute heart disease). The problem with these alternatives is first that they somewhat beg the question (presumably the person telling us what cues to look at must have learned the importance at some point), but more worryingly they reduce the precision of the heuristics. If the “best” in TTB no longer only refers to cues with the highest ecological validity but can mean any subjective rank-ordering of cues—be it one provided by evolution, social learning, the idiosyncratic (and often erroneous) learning of an individual (Gigerenzer et al., 2008), or even the “environmental accessibility” of cues (Gigerenzer & Brighton, 2009, p. 129), then the precision of the heuristic is lost (cf., Dougherty, Thomas, et al., 2008; Newell, 2011; but see Katsikopoulos, Schooler, & Hertwig, 2010).

The testability of the approach can be critiqued on the grounds that the process by which different heuristics are selected for use in particular environments remains unspecified. Gigerenzer and Brighton (2009) suggest that selection occurs via a “mostly unconscious process” (p. 129) but go on to describe three potential selection mechanisms: constraints of memory, constraints of the environment, and feedback on the performance of heuristics.

How does the state of memory at the time of judgment dictate heuristic choice? Gigerenzer and Brighton suggest that if you are asked to predict who of two players will win a tennis match at Wimbledon and you have heard of one but not the other, you “select” the recognition heuristic and predict that the recognized player will win. If you have heard of both but the name of one came to mind more quickly than the other, then you use the *fluency heuristic* (Schooler & Hertwig, 2005) and predict that the more fluently processed player will win. Finally, if you know both players and can recall some facts about each (e.g., their rankings, previous performance at Wimbledon), then memory “selects” TTB, and you choose the player pointed to by the most valid discriminating cue. The environmental constraints mesh with the memory ones to ensure that only heuristics that are ecologically rational are selected. The recognition heuristic is useful in predicting tennis tournament winners because the correlation between recognition and winners is strong and positive; if the correlation had been low (or even negative), then the environment/memory would have “selected” a different heuristic.

Such descriptions of how the selection of heuristics occurs make intuitive sense, but how different are they from the ideas of natural assessment and attribute substitution? One can describe fast and frugal heuristics according to the implementation of particular rules, but the underlying processes seem remarkably similar. When faced with a judgment about a target attribute (e.g., How likely is it that Andrew Murray will beat Roger Federer?), one *substitutes* it with information provided through natural assessments of recognition, fluency, or cue-based knowledge. Certainly these heuristic models can be implemented and tested in computer programs (e.g., Schooler & Hertwig, 2005), but this does not privilege them; other plausible and accurate models of human judgment that do not require recourse to the selection of discrete heuristics from an adaptive toolbox can also be represented as computational models (e.g., Chater et al. 2003; Dougherty, Gettys, & Ogden, 1999; Dougherty, Franco-Watkins, et al., 2008; Juslin & Persson, 2002; Lee & Cummins, 2004; Lee & Newell, 2011; Newell & Lee, 2011).

p. 612

The third selection mechanism—feedback—has been emphasized in recent computational models of strategy selection. Rieskamp and colleagues (Rieskamp, 2008; Rieskamp & Otto, 2006) demonstrate how principles of reinforcement learning can explain how people learn to adopt (and adapt) strategies that search for and integrate more or fewer cues depending on the structure of the environment. Although, again, the adaptive toolbox model is not privileged in being able to explain such adaptation—(see Lee & Cummins, 2004; Newell, 2005; Newell & Lee, 2011; for an alternative framework based on evidence-accumulation models)—attempting to explain how people adapt judgment behavior is a useful advance on a problem that has plagued the literature for a long time (cf., Bröder & Newell, 2008; Payne, Bettman, & Johnson, 1993). The approach also emphasizes the key role for learning from feedback—something that was often absent in the heuristics and biases approach due to its focus on one-shot judgments (cf., Newell et al., 2007).

Looking Inside, Outside, and Ahead

The fast and frugal approach has had a positive effect on the drive to “take the interface between mind and environment seriously” (Fiedler & Juslin, 2006, p. 9). This renewed emphasis on the ecology of judgment has brought the ideas of Brunswik (1956) and Simon (1955) back into vogue and also offers a way to reconcile the uneasy juxtaposition of the numerous studies that exhibit impressive judgment accuracy (e.g., Karelaia & Hogarth, 2008) and dramatic errors and biases (e.g., Tversky & Kahneman, 1974). By thinking carefully about how information presented to us in the world “outside” influences the representation of information “inside” our minds, a deeper understanding of the judgment process can be gained (e.g., Kahneman & Klein, 2009).

The *naïve* intuitive statistician approach (Fiedler, 2000; Fiedler & Juslin, 2006) focuses directly on this interface between mind and environment. The metaphor echoes the historical view (mentioned earlier) that statistical models provided good approximations for psychological theories of judgment (the *intuitive statistician* metaphor; Peterson & Beach, 1967). The novel insight is to argue that when judgments go awry it

is not because the internal cognitive mechanisms *per se* are at fault; rather, the judge is naïve with respect to the samples that provide the input for these mechanisms. These *biased samples* may be created via bias in the external environment (e.g., the heightened media coverage of homicide over suicide, Lichtenstein et al., 1978), bias in memory representations (the letter “k” effect, Tversky & Kahneman, 1974), or even bias induced via the use of unrepresentative experimental materials (such as in the overconfidence phenomena in general knowledge; see Juslin, 1994).

The important point is that while previous explanations of these kinds of effects have appealed to cognitive algorithms that sometimes lead us astray (“availability”), the sampling explanation shifts the onus of explanation to the property of the sample. Thus, biased judgments are not due to faulty metacognition about ease of recall; the deficit in metacognition is in the inability to correct for the bias in the input. When more representative samples are used, such as using a large sample of letters rather than only the five consonants that *do* appear more often in the third than first position of words, “availability biases” like the “letter k” effect can be shown to disappear (Sedlmeier, Hertwig, & Gigerenzer, 1998).

Rethinking the Sample: Inside and Outside Views on Judgment

The rhetoric of the naïve intuitive statistician approach implies that judgment should be improved by increasing awareness of the potential bias in external samples of information. This may be difficult to achieve. Fiedler (2000) gives the example of an expert estimating the causal impact of alcohol (of a certain concentration) on traffic accidents. Although the expert may be fully aware that the available samples are conditional on the criterion (i.e., the occurrence of an accident), taking this into account when quantifying an estimate—perhaps via Bayes’s theorem—may not be possible because the base rate $p(\text{alcohol})$ and the false alarm rate $p(\text{alcohol}/\text{no accident})$ are unknown.

Although it may sometimes not be possible to find necessary information in the external environment, careful thought about the representation of information in our minds can help to overcome erroneous judgments. Consider a judgment about how long it will take you to write an essay (Buehler, Griffin, & Ross, 1994). You could consider the qualities of the particular essay (e.g., complexity of the topic, word limit, etc.) or you could consider the essay in the context of other similar ones that you have written and the time that they took. Kahneman and Lovalló (1993) label these two approaches as “inside” and “outside” views, respectively. Here, the inside and outside refer not to mind and environment, but rather to ways of constructing the representation of information in the mind. Kahneman and Lovalló argue that although an “outside view” is often nonintuitive, adopting one diminishes the types of “planning fallacies” that occur when the specifics of a single case are relied upon (Kahneman & Lovalló, 1993). In sampling terms, the outside view promotes the gathering of a set of instances, but with the caveat that the judge needs to be sensitive to how the instances are generated. A similar idea is found in the advice to “consider the opposite” (Larrick, 2004) when making judgments. Simply asking why an initial judgment might be wrong, or whether you should change some of your assumptions before committing to an answer, has been shown to be effective in overcoming anchoring biases in numerical estimation tasks (e.g., Mussweiler, Strack, & Pfeiffer, 2000) and improving estimates in general knowledge tasks (Herzog & Hertwig, 2009).

Are There Two Systems Underlying Judgment?

This general notion that sometimes “more thinking” about a judgment can alleviate errors is sometimes discussed in terms of the competition between two distinct systems. The typical contrast is between a “System 1,” depicted as intuitive or associative, and a “System 2,” which is analytic, reflective, or rule-based (e.g., Evans, 2008; Kahneman, 2003; Kahneman & Frederick, 2002; Sloman, 1996; Stanovich & West, 2000). The basic idea is that initial, intuitive answers are proposed by System 1 (e.g., “Linda must be a feminist bank teller”) but are simultaneously monitored by System 2 and, if necessary, (and circumstances permit) can be overridden (e.g., “She can’t be because that violates a basic law of probability”). This characterisation leads to important questions about how such monitoring happens and what determines which system we “listen to” in giving our final response (i.e., What are the circumstances that permit endorsement, correction, or the overriding of System 1 by System 2?)

These are not trivial questions, and indeed a large literature has evolved attempting to identify when System 1 or 2 processes dominate (e.g., Evans, 2008; Kahneman, 2003). This literature highlights factors such as the time available or taken for a decision (e.g., DeNeys & Glumicic, 2008), the availability of cognitive resources

(DeNeys, 2006), individual differences in intelligence (e.g., Stanovich & West, 2000), and the effect of instructions encouraging people to think “logically” (e.g., Bonner & Newell, 2010).

Despite the voluminous research and the obvious allure of the dual-system framework, there is debate about the explanatory role that such a framework plays. The advantages of describing processes and heuristics as under the operation of System 1 and System 2 are, some argue, outweighed by the disadvantages for theoretical progress. A key problem in the dual-system literature is the plethora of often vague and imprecise terms used to describe the characteristics of the systems, and a failure to specify the manner in which systems interact and/or exchange information. This has led, arguably, to an illusion of convergence whereby researchers appear to be describing similar models and frameworks that, in fact, differ in many crucial aspects (Evans, 2008). This tendency could impede progress on understanding the phenomena of interest through a (potentially misguided) desire to dichotomize (e.g., Bonner & Newell, 2010; Gigerenzer & Regier, 1996; Keren & Schul, 2009; Marewski, Gaissmaier, & Gigerenzer, 2010; see also Newell, Dunn, & Kalish, 2011, for a related discussion in category learning research).

Nevertheless, some argue that even imprecise frameworks such as those offered by dual-process protagonists are useful because of their potential to guide search for commonalities in processes, identify analogies across domains, and prevent overly narrow interpretations of phenomena (e.g., Kahneman, 2003, 2011). Time will tell whether such optimism is warranted; what is clear is that the development of dual-process accounts will be keenly watched and debated in the coming years.

A Causal Framework: Integrating Coherence and Correspondence?

p. 614

Whatever the eventual outcome of the dual-system debate, it is clear from the path followed in this chapter through traditions that have emphasized coherence and correspondence criteria that any successful approach will need to apply both to understand judgment. A novel approach that has the potential to achieve this integration is one which incorporates a role for causal knowledge. ↪ Krynski and Tennenbaum (2007) suggest that deviations from normative standards, such as that exemplified by base rate neglect, can be explained via a mismatch between the statistics provided to participants and the intuitive causal models that participants construct when attempting to answer probability judgment problems. Krynski and Tennenbaum argue that when faced with a problem a participant engages in three steps of reasoning. First, causal domain knowledge is used to construct a causal model of the situation; second, the available statistical data (the probabilities presented in the problem) are used to set parameter values on the model, and finally judgments are made via Bayesian inference over the parameterized model. Krynski and Tennenbaum demonstrated that when inferences were made over intuitive causal models, base-rate neglect was reduced relative to problems in which the models were nonintuitive (e.g., a medical diagnosis task in which false-positive rates had to be included).

The approach taken by Krynski and Tennenbaum (2007) is consistent with a current popular view that emphasizes the potential of a causal Bayesian framework for understanding cognition (e.g., Chater & Oaksford, 2008). The approach provides a possible way to incorporate correspondence and coherence criteria into our explanations of judgment behavior. Coherence standards are readily accommodated via the prescription of an optimal standard—Bayes’s rule—for inference. The departure from earlier research focusing on Bayes (e.g., Edwards, 1968) is the inclusion of causal mental models (Pearl, 2000) that specify and typically reduce the number of parameters over which the inference needs to be computed. Correspondence criteria are incorporated via the inclusion of causal knowledge in the models. As Krynski and Tennenbaum note, “causal Bayesian reasoning is often more ecologically adaptive, because it can leverage available causal knowledge to make appropriate judgments even when there is not sufficient statistical data available to make rational inferences under noncausal Bayesian norms” (p. 434). Thus, the causal Bayesian framework can combine coherence via Bayesian inference and correspondence via causal models (or priors) about the facts of the world to provide largely accurate judgments. The brief sketch provided here can only indicate the potential, but maybe the causal Bayesian framework can capitalize on the inseparability of coherence and correspondence and further elucidate how we make judgments under uncertainty.

Psychological investigations of judgment under uncertainty have proved fertile ground over the past 50 years. This chapter viewed key findings from this rich tradition through the lens of correspondence and coherence criteria for appraising judgment. The research highlights the inseparability of the two criteria and the need to consider both in developing a fully fledged account of judgment. The research appears to be heading in an exciting and fruitful direction—a direction in which the properties of tasks and the constraints of the external environment are given their deserved prominence (e.g., Edwards, 1983; Fiedler & Juslin, 2006) and are combined with appropriate rational benchmarks for assessing judgment (e.g., Anderson, 1990; Chater & Oaksford, 2008; Juslin, Nilsson, & Winman, 2009; Krynski & Tennenbaum, 2007; McKenzie, 2005). With luck, and good judgment, continuing on this promising path will lead us to discover some of things we do not know about judgment under uncertainty (even if we do not yet know what we do not know).

Future Directions

1. Why is some information more accessible and thus more readily relied upon during the judgment process? (See Shah & Oppenheimer, 2009, for recent work on this topic.)
2. Do we need to posit multiple discrete heuristics to explain judgment, or will a domain-general perspective prove more fruitful?
3. Can we develop a taxonomy of tasks to examine the interaction of task and individual variables and better relate coherence and correspondence criteria? (See Juslin, Nilsson, & Winman, 2009, for a fascinating exploration of how this might be achieved.)
4. What is the potential of the causal Bayesian framework for explaining judgment under uncertainty?

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